

COS1501

UNIT 4

PROOFS INVOLVING SETS

Symbols and proof language first - then complete guide-level theory
Every main topic includes a medium and a hard fully worked example

Colour key

TEAL definitions and theorems	BLUE medium worked examples	PURPLE hard worked examples	RED / ORANGE traps and exam notes
----------------------------------	-----------------------------------	--------------------------------	--------------------------------------

Part A - Symbols used in Unit 4 proofs

A proof is a chain of reversible or one-way statements. The symbols below are not decoration: each one tells you what logical move is allowed.

EXAM NOTE - Read complete statements

In $x \in A \Rightarrow x \in B$, the symbol $\in$ describes membership and $\Rightarrow$ describes the logical direction. Do not read the line as loose algebra.

SYMBOL	TERMINOLOGY / READ AS	MEANING IN A PROOF	BRIEF EXAMPLE
$\in$	is an element of	Starts or ends an element-chasing step.	$x \in A \cap B$
$\notin$	is not an element of	Expresses failed membership.	$x \notin A \cup B$
$\subseteq$	is a subset of	Every left-set element lies in the right set.	$A \cap B \subseteq A$
$\subset$	is a proper subset of	Subset and unequal.	$\{1\} \subset \{1, 2\}$
$=$	is equal to	The sets have exactly the same elements.	$A \cup B = B \cup A$
$\neq$	is not equal to	At least one element separates the sets.	$A - B \neq B - A$ in general
$\forall$	for all	Introduces a universal assertion.	$\forall A \subseteq U, A \cup \emptyset = A$
$\exists$	there exists	Introduces at least one witness.	$\exists x \in A - B$

Proof symbols - logical direction and completion

SYMBOL	TERMINOLOGY / READ AS	MEANING IN A PROOF	BRIEF EXAMPLE
$\Rightarrow$	implies / only if	Truth on the left guarantees truth on the right.	$x \in A \cap B \Rightarrow x \in A$
$\Leftarrow$	is implied by / if	The right statement guarantees the left.	$x \in A \Leftarrow x \in A \cap B$
$\Leftrightarrow$	if and only if	Both directions hold; each statement is equivalent.	$x \in A' \Leftrightarrow x \notin A$
iff	if and only if	Word abbreviation for $\Leftrightarrow$.	$A=B$ iff $A \subseteq B$ and $B \subseteq A$
LHS	left-hand side	Expression before the equality sign.	$\text{LHS} = A - (B \cup C)$
RHS	right-hand side	Expression after the equality sign.	$\text{RHS} = (A - B) \cap (A - C)$
QED	quod erat demonstrandum	Marks the completed proof.	Therefore $A=B$. QED
$\neg$	not / negation	Reverses a truth condition.	$\neg(x \in A)$ means $x \notin A$

Set-operation and counting symbols

SYMBOL	TERMINOLOGY / READ AS	MEANING IN A PROOF	BRIEF EXAMPLE
U	universal set	The domain containing every set in the argument.	$A, B, C \subseteq U$
$\emptyset$	empty set	The set with no elements.	$A \cap A' = \emptyset$
U	union / inclusive or	Membership in at least one set.	$x \in A \cup B$ iff $x \in A$ or $x \in B$
n	intersection / and	Membership in both sets.	$x \in A \cap B$ iff $x \in A$ and $x \in B$
–	difference	Left set with right-set members removed.	$x \in A - B$ iff $x \in A$ and $x \notin B$
A'	complement	Members of U outside A.	$x \in A'$ iff $x \notin A$
+	symmetric difference	Membership in exactly one set.	$A + B = (A - B) \cup (B - A)$
 A 	cardinality	Number of distinct elements in finite A.	$ \{a, b\} = 2$
P(A)	power set	The set of all subsets of A.	$X \in P(A)$ iff $X \subseteq A$

Part B - Terminology you must use correctly

These words describe different mathematical jobs. Mixing them up is one of the fastest ways to lose proof marks.

TERM	PRECISE MEANING
Venn diagram	A region picture of subsets inside a universal set.
shaded region	The part representing the result of a set expression.
set equality	$A=B$: the two specific sets have exactly the same members.
set identity	An equality true for every allowed choice of its set variables.
universal assertion	A statement claiming something for all allowed sets or elements.
counterexample	One allowed choice that makes a universal assertion false.
arbitrary element	An element chosen with no special property beyond stated membership.
witness	A specific element or set establishing existence or failure.
mutual containment	Proving $LHS \subseteq RHS$ and $RHS \subseteq LHS$ to conclude equality.
converse	The reverse implication or second containment direction.
iff chain	A chain of logically equivalent membership statements.
element chasing	Translating set membership step by step using definitions.
inclusion-exclusion	A counting rule correcting overlap counted more than once.
sum rule	For disjoint finite sets, cardinalities add directly.
specific-set proof	A proof using the actual admission rules of named concrete sets.
QED	Traditional marker that the stated claim has been proved.

Unit roadmap and required proof standard

Unit 4 moves from calculating sets to proving statements about all sets. A diagram can suggest a result, but a formal proof must translate membership definitions accurately and justify every direction.

EXAM NOTE - Guide-level outcomes

Construct and interpret Venn diagrams; test set equations; prove identities using mutual containment and iff chains; prove subset claims; disprove universal claims with counterexamples; use inclusion-exclusion; and prove equalities for specific algebraic sets.

PART	FOCUS	COMPLETE COVERAGE
1	Venn language	regions, shading, staged construction, limitations
2	Identity proofs	mutual containment, iff chains, De Morgan and distribution
3	Proof decisions	subset proofs, identities, false claims and counterexamples
4	Counting proofs	inclusion-exclusion, sum rule, two- and three-set problems
5	Specific sets	algebraic admission rules and exact equivalence proofs

- Can I distinguish evidence from a full proof?
- Can I prove both directions of equality without skipping definitions?
- Can I design a counterexample that targets the exact region where two expressions differ?
- Can I separate inclusive intersections from pair-only regions in counting problems?

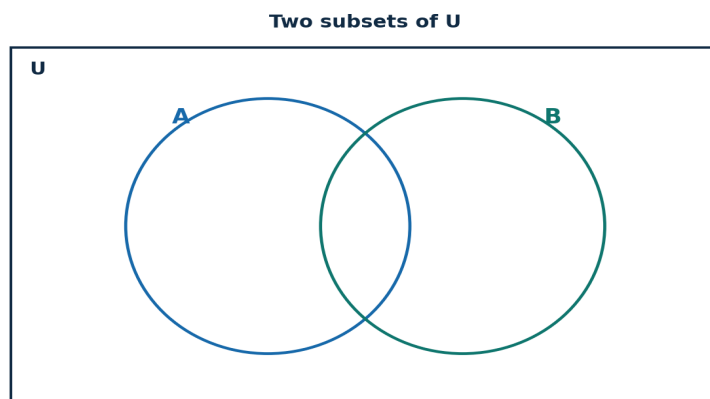
1. Anatomy of a Venn diagram

DEFINITION - Anatomy of a Venn diagram

A Venn diagram represents U by a rectangle and each subset by a closed curve inside U .

Every point in a region represents a possible element with that membership pattern. Overlap means simultaneous membership.

The shaded region is the result set, not merely a visual emphasis. Labels and U must be shown.



EXAM NOTE - Diagram discipline

A picture is only useful when all relevant sets and the universal set appear and the intended result is clearly shaded.

MEDIUM WORKED EXAMPLE - Place elements in regions

Problem. Let $U = \{1, 2, 3, 4, 5\}$, $A = \{1, 2, 3\}$, $B = \{3, 4\}$. Place each element conceptually.

Solution.

1 and 2 lie in A only; 3 lies in $A \cap B$; 4 lies in B only; 5 lies outside $A \cup B$.

HARD WORKED EXAMPLE - Infer conditions from regions

Problem. An element x lies inside A and outside B. Which statements are forced?

Solution.

$x \in A$, $x \notin B$, $x \in A - B$, $x \in A \cup B$, $x \notin A \cap B$ and $x \in B'$. No conclusion about a third set C is possible.

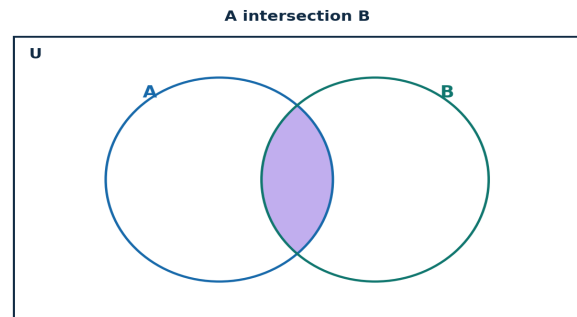
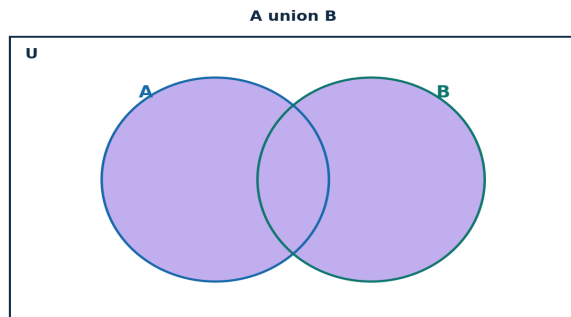
2. Union and intersection shading

DEFINITION - Union and intersection shading

Union shades every region belonging to A or B or both; intersection shades only the overlap.

Translate before shading: $x \in A \cup B$ iff $x \in A$ or $x \in B$; $x \in A \cap B$ iff $x \in A$ and $x \in B$.

Common elements are not counted or shaded twice; a set contains distinct elements.



COMMON TRAP - Inclusive or

The “or” in union includes elements belonging to both sets.

MEDIUM WORKED EXAMPLE - Read the shading

Problem. For $A = \{1, 2, 4\}$ and $B = \{2, 3, 4\}$, list $A \cup B$ and $A \cap B$.

Solution.

$A \cup B = \{1, 2, 3, 4\}$; $A \cap B = \{2, 4\}$.

HARD WORKED EXAMPLE - Recover sets from regions

Problem. Suppose $A\text{-only} = \{a, c\}$, $A \cap B = \{d, e\}$, $B\text{-only} = \{f\}$, and $\text{outside} = \{g\}$. Find A, B, $A \cup B$ and $A \cap B$.

Solution.

$A = \{a, c, d, e\}$; $B = \{d, e, f\}$; $A \cup B = \{a, c, d, e, f\}$; $A \cap B = \{d, e\}$.

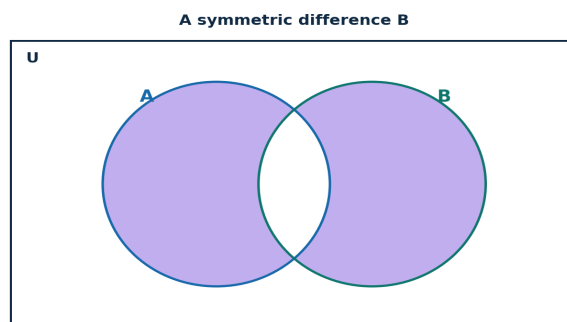
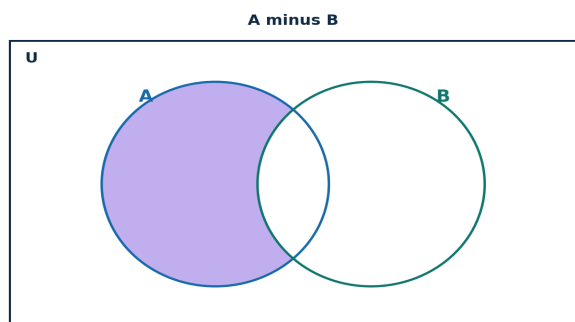
3. Complement, difference and symmetric difference

DEFINITION - Complement, difference and symmetric difference

Complement selects outside a set; difference selects the left-only region; symmetric difference selects both one-set-only regions.

$$A' = U - A; A - B = A \cap B'; A + B = (A - B) \cup (B - A)$$

Symmetric difference is exclusive or. Difference is directional; complement depends on U.



COMMON TRAP - Three distinct ideas

A' may include elements outside both A and B; $A - B$ stays inside A; $A + B$ stays inside $A \cup B$ and deletes the overlap.

MEDIUM WORKED EXAMPLE - Compute three operations

Problem. Let $U = \{1, 2, 3, 4, 5, 6\}$, $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5\}$. Find A' , $A - B$ and $A + B$.

Solution.

$A' = \{5, 6\}$; $A - B = \{1, 2\}$; $A + B = \{1, 2, 5\}$.

HARD WORKED EXAMPLE - Reconstruct B

Problem. Let $U = \{1, 2, 3, 4, 5\}$, $A = \{1, 2, 3\}$, and $A + B = \{2, 4, 5\}$. Find B.

Solution.

Membership is toggled exactly at 2, 4, 5. Thus $B = \{1, 3, 4, 5\}$.

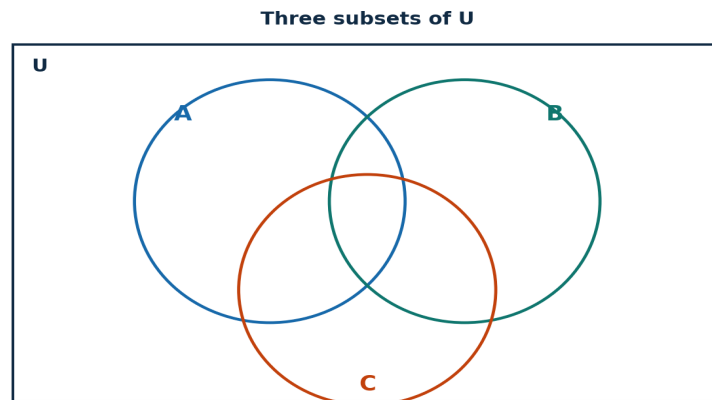
4. Three-set regions

DEFINITION - Three-set regions

Three overlapping sets create eight elementary regions: seven inside $A \cup B \cup C$ and one outside all three.

The central region belongs to all three. Each pairwise intersection includes the centre unless “only” is stated.

When shading, determine the truth of the expression in each elementary region.



EXAM NOTE - Pairwise intersection is inclusive

$A \cap B$ contains both the A-and-B-only region and $A \cap B \cap C$.

MEDIUM WORKED EXAMPLE - Classify one region

Problem. If $x \in A \cap C$ and $x \notin B$, decide $x \in A \cap (B + C)$.

Solution.

Since $x \notin B$ and $x \in C$, x belongs to exactly one of B, C , so $x \in B + C$. With $x \in A$, the statement is true.

HARD WORKED EXAMPLE - Count all region types

Problem. How many elementary regions arise from three sets, and why?

Solution.

Each element has two choices for each of A, B, C : in or out. Thus $2^3 = 8$ membership patterns, including the outside region.

5. Staged shading of complex expressions

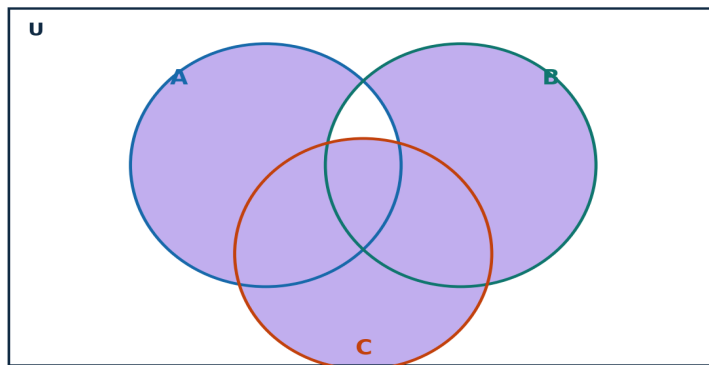
DEFINITION - Staged shading of complex expressions

Build a complex Venn result from the innermost brackets outward.

For $[(A \cup B) - (A \cap B)] \cup C$, first form $A \cup B$, then remove $A \cap B$, then add all of C .

The intermediate result $(A \cup B) - (A \cap B)$ is $A + B$.

$[(A \text{ union } B) \text{ minus } (A \text{ intersection } B)] \text{ union } C$



EXAM NOTE - Do not shade by guessing

Write the intermediate expressions in order. A correct final picture with unexplained guessing is fragile under exam pressure.

MEDIUM WORKED EXAMPLE - Simplify before drawing

Problem. Simplify $[(A \cup B) - (A \cap B)] \cup C$.

Solution.

The bracket is the symmetric difference $A + B$. Therefore the expression is $(A + B) \cup C$.

HARD WORKED EXAMPLE - Region membership test

Problem. For $E = [(A \cup B) - (A \cap B)] \cup C$, determine whether an element in all three sets belongs to E and whether an A-only element belongs.

Solution.

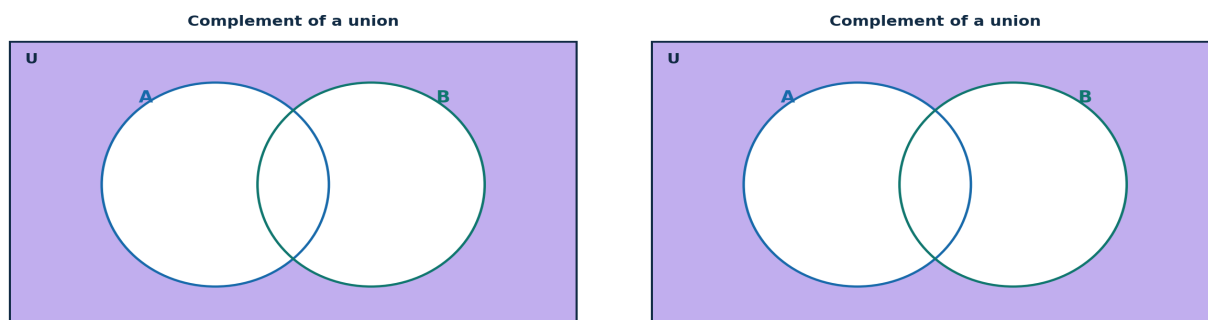
A triple member is removed from $A \cup B$ by subtracting $A \cap B$, but added back because it belongs to C ; so it is in E . An A-only element survives the difference and is also in E .

6. Using Venn diagrams to test identities

DEFINITION - Using Venn diagrams to test identities

Two expressions appear equal when their final shaded regions coincide for every elementary region.

Draw the same set arrangement for both sides. Shade each side independently and compare all regions. If regions differ, target one differing region to construct a counterexample.



EXAM NOTE - Evidence versus proof

Matching diagrams strongly indicate an identity, but the guide then expects a formal element proof when the claim appears true.

MEDIUM WORKED EXAMPLE - Test De Morgan visually

Problem. Compare $(A \cup B)'$ with $A' \cap B'$.

Solution.

Both shade exactly the region outside A and outside B. The diagram suggests equality.

HARD WORKED EXAMPLE - Find a visual failure

Problem. Compare $A - (B - C)$ with $(A - B) - C$ using region logic.

Solution.

Take $x \in A \cap C$ with $x \notin B$. Then $x \notin B - C$, so x belongs to $A - (B - C)$. But $x \in C$, so it is removed from $(A - B) - C$. The expressions are not identical.

7. Limits of Venn-diagram proofs

DEFINITION - Limits of Venn-diagram proofs

Venn diagrams become unwieldy beyond three sets and can hide assumptions about empty or coincident regions.

A generic drawing usually shows every region as non-empty even though actual sets may have empty regions.

Formal element-chasing avoids dependence on scale, shape or visual interpretation.

NOTE - Use the diagram correctly

Use it to investigate, organise regions and design a counterexample; use definitions for the final rigorous argument.

MEDIUM WORKED EXAMPLE - Empty-set edge case

Problem. Does the identity $A \cap A' = \emptyset$ remain true when $A = \emptyset$?

Solution.

Yes. Then $A' = U$ and $\emptyset \cap U = \emptyset$.

HARD WORKED EXAMPLE - Why pictures can mislead

Problem. A drawing shows A and B overlapping. Does that prove arbitrary A and B are not disjoint?

Solution.

No. The circles show possible membership regions, not a claim that each region contains an element. Actual $A \cap B$ may be empty.

8. Equality proofs by mutual containment

DEFINITION - Equality proofs by mutual containment

To prove $LHS=RHS$, prove $LHS \subseteq RHS$ and $RHS \subseteq LHS$.

First choose an arbitrary x in the left side and derive right-side membership. Then reverse direction with arbitrary x in the right side.

The element must be arbitrary: a single successful example proves nothing universal.

EXAM NOTE - Equality standard

Both containments are necessary unless one iff chain establishes both at once.

MEDIUM WORKED EXAMPLE - Prove commutativity of union

Problem. Prove $A \cup B = B \cup A$.

Solution.

If $x \in A \cup B$, then $x \in A$ or $x \in B$, hence $x \in B$ or $x \in A$, so $x \in B \cup A$. The reverse containment is identical with A, B exchanged. Therefore equality holds.

HARD WORKED EXAMPLE - Prove absorption

Problem. Prove $A \cup (A \cap B) = A$ by mutual containment.

Solution.

If $x \in A \cup (A \cap B)$, then $x \in A$ or $(x \in A \text{ and } x \in B)$, so $x \in A$. Conversely, $x \in A$ implies $x \in A \cup (A \cap B)$. Thus both containments hold.

9. Iff chains

DEFINITION - Iff chains

An iff chain proves equality by showing x belongs to the left set exactly when it belongs to the right set.

Every link must be reversible. A one-way implication cannot be inserted into an iff chain.

Finish by invoking extensionality: equal membership for every x means equal sets.

COMMON TRAP - Check reversibility

At each “iff”, ask whether the statement below also guarantees the statement above.

MEDIUM WORKED EXAMPLE - Double complement

Problem. Prove $(A')' = A$.

Solution.

For arbitrary $x \in U$: $x \in (A')'$ iff $x \notin A'$ iff it is not true that $x \notin A$ iff $x \in A$. Therefore the sets are equal.

HARD WORKED EXAMPLE - Symmetric difference commutes

Problem. Prove $A+B=B+A$ with an iff chain.

Solution.

$x \in A+B$ iff $(x \in A \text{ and } x \notin B) \text{ or } (x \in B \text{ and } x \notin A)$ iff $(x \in B \text{ and } x \notin A) \text{ or } (x \in A \text{ and } x \notin B)$ iff $x \in B+A$.

10. Associativity of union

DEFINITION - Associativity of union

Union is associative: $A \cup (B \cup C) = (A \cup B) \cup C$.

Associativity changes grouping, not order. Both sides mean membership in at least one of A,B,C.

$x \in A \cup (B \cup C)$ iff $x \in A$ or $(x \in B$ or $x \in C)$

NOTE - Do not confuse laws

Commutativity swaps order; associativity moves brackets.

MEDIUM WORKED EXAMPLE - Complete the proof

Problem. Show $A \cup (B \cup C) \subseteq (A \cup B) \cup C$.

Solution.

If $x \in A \cup (B \cup C)$, then $x \in A$ or $(x \in B$ or $x \in C)$. Regrouping “or” gives $(x \in A$ or $x \in B)$ or $x \in C$, hence $x \in (A \cup B) \cup C$.

HARD WORKED EXAMPLE - Prove the full identity

Problem. Prove associativity in one iff chain.

Solution.

$x \in A \cup (B \cup C)$ iff $x \in A$ or $(x \in B$ or $x \in C)$ iff $(x \in A$ or $x \in B)$ or $x \in C$ iff $x \in (A \cup B) \cup C$. This holds for every x .

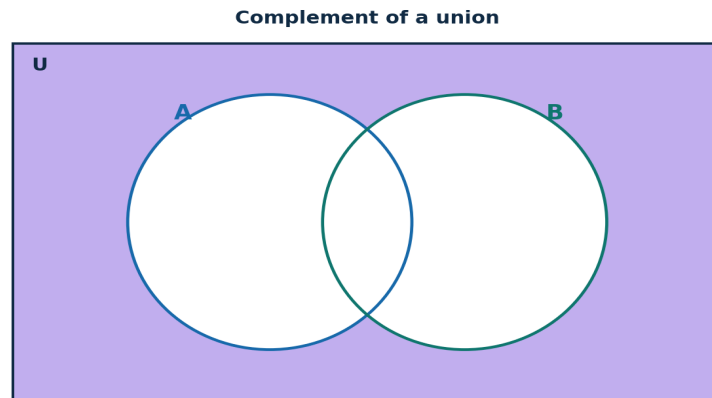
11. De Morgan law for a union

DEFINITION - De Morgan law for a union

The complement of a union equals the intersection of the complements: $(A \cup B)' = A' \cap B'$.

Not being in $A \cup B$ means being in neither A nor B . Negation changes “or” to “and”.

$x \notin A \cup B$ iff $x \notin A$ and $x \notin B$



COMMON TRAP - Negation changes the connector

It is wrong to write $x \notin A \cup B$ iff $x \notin A$ or $x \notin B$. One failed membership is not enough to stay outside the union.

MEDIUM WORKED EXAMPLE - Numerical verification

Problem. Let $U = \{1, 2, 3, 4, 5\}$, $A = \{1, 2\}$, $B = \{2, 3\}$. Verify $(A \cup B)' = A' \cap B'$.

Solution.

$A \cup B = \{1, 2, 3\}$, so its complement is $\{4, 5\}$. Also $A' = \{3, 4, 5\}$, $B' = \{1, 4, 5\}$, and their intersection is $\{4, 5\}$.

HARD WORKED EXAMPLE - Formal iff proof

Problem. Prove $(A \cup B)' = A' \cap B'$.

Solution.

$x \in (A \cup B)'$ iff $x \notin A \cup B$ iff $x \notin A$ and $x \notin B$ iff $x \in A'$ and $x \in B'$ iff $x \in A' \cap B'$.

12. De Morgan law for an intersection

DEFINITION - De Morgan law for an intersection

The complement of an intersection equals the union of the complements: $(A \cap B)' = A' \cup B'$.

Not being in both sets means failing at least one membership. Negation changes “and” to “or”.

$$x \notin A \cap B \text{ iff } x \notin A \text{ or } x \notin B$$

NOTE - Outside the intersection

An A-only element is outside $A \cap B$ even though it belongs to A.

MEDIUM WORKED EXAMPLE - Numerical verification

Problem. Let $U = \{1, 2, 3, 4\}$, $A = \{1, 2\}$, $B = \{2, 3\}$. Verify the law.

Solution.

$A \cap B = \{2\}$, so $(A \cap B)' = \{1, 3, 4\}$. $A' = \{3, 4\}$, $B' = \{1, 4\}$; their union is $\{1, 3, 4\}$.

HARD WORKED EXAMPLE - Formal iff proof

Problem. Prove $(A \cap B)' = A' \cup B'$.

Solution.

$$x \in (A \cap B)' \text{ iff } x \notin A \cap B \text{ iff } x \notin A \text{ or } x \notin B \text{ iff } x \in A' \text{ or } x \in B' \text{ iff } x \in A' \cup B'.$$

13. Difference across a union

DEFINITION - Difference across a union

$$A - (B \cup C) = (A - B) \cap (A - C).$$

An element survives removal of $B \cup C$ exactly when it is in A , outside B and outside C .

$$A - (B \cup C) = A \cap (B \cup C)' = A \cap B' \cap C'$$

NOTE - Repeated A is harmless

In $(A - B) \cap (A - C)$, the condition $x \in A$ occurs twice but remains one condition.

MEDIUM WORKED EXAMPLE - Check concrete sets

Problem. Let $A = \{1, 2, 3, 4\}$, $B = \{2, 5\}$, $C = \{3, 5\}$. Verify the identity.

Solution.

$$\text{LHS} = A - \{2, 3, 5\} = \{1, 4\}. \text{ RHS} = \{1, 3, 4\} \cap \{1, 2, 4\} = \{1, 4\}.$$

HARD WORKED EXAMPLE - Prove by iff

Problem. Prove $A - (B \cup C) = (A - B) \cap (A - C)$.

Solution.

$$x \in A - (B \cup C) \text{ iff } x \in A \text{ and } x \notin B \cup C \text{ iff } x \in A, x \notin B \text{ and } x \notin C \text{ iff } x \in A - B \text{ and } x \in A - C \text{ iff } x \in (A - B) \cap (A - C).$$

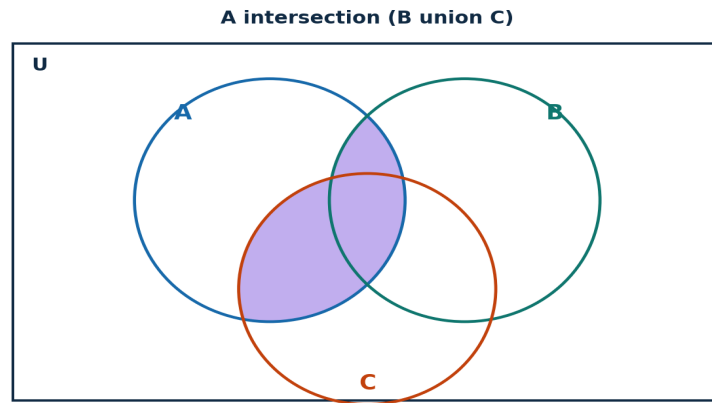
14. Distributivity of intersection

DEFINITION - Distributivity of intersection

Intersection distributes over union: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

Membership means x is in A and also in at least one of B, C . Logical distribution gives the right side.

p and $(q$ or $r)$ iff $(p$ and $q)$ or $(p$ and $r)$



COMMON TRAP - Keep the repeated condition

Both branches on the right must contain $x \in A$.

MEDIUM WORKED EXAMPLE - Verify with finite sets

Problem. Let $A = \{1, 2, 3\}$, $B = \{2, 4\}$, $C = \{1, 4\}$. Compute both sides.

Solution.

$B \cup C = \{1, 2, 4\}$; intersect A to get $\{1, 2\}$. Also $A \cap B = \{2\}$, $A \cap C = \{1\}$; their union is $\{1, 2\}$.

HARD WORKED EXAMPLE - Prove distribution

Problem. Give an iff proof.

Solution.

$x \in A \cap (B \cup C)$ iff $x \in A$ and $(x \in B$ or $x \in C)$ iff $(x \in A$ and $x \in B)$ or $(x \in A$ and $x \in C)$ iff $x \in (A \cap B) \cup (A \cap C)$.

15. Proving subset statements

DEFINITION - Proving subset statements

To prove $S \subseteq T$, choose arbitrary $x \in S$ and derive $x \in T$. Only one direction is required.

Do not attempt a reverse direction unless equality is claimed. A subset proof may end while T still contains additional elements.

To disprove $S \subseteq T$, find $x \in S - T$.

EXAM NOTE - Proof target

Write the exact target membership before manipulating: "We must show $x \in T$."

MEDIUM WORKED EXAMPLE - Simple containment

Problem. Prove $A \cap B \subseteq A \cup B$.

Solution.

Let $x \in A \cap B$. Then $x \in A$ and $x \in B$, hence $x \in A$ or $x \in B$. Therefore $x \in A \cup B$.

HARD WORKED EXAMPLE - Power-set containment

Problem. Prove $P(A) \cup P(B) \subseteq P(A \cup B)$.

Solution.

Let $X \in P(A) \cup P(B)$. Then $X \subseteq A$ or $X \subseteq B$. In either case every member of X lies in $A \cup B$, so $X \subseteq A \cup B$; hence $X \in P(A \cup B)$.

16. Identity versus equation

DEFINITION - Identity versus equation

A set identity is true for every allowed choice of its set variables; an ordinary equality may hold only for particular sets.

$A \cup B = B \cup A$ is an identity. $A - B = B - A$ is not, although it happens to hold for some choices such as $A = B$.

One counterexample destroys identity status; many successful examples cannot establish it.

EXAM NOTE - Quantifiers matter

"For all A, B " is defeated by one failure. "For these A, B " requires calculation only for the named sets.

MEDIUM WORKED EXAMPLE - Classify a statement

Problem. Is $A \cap B = A$ an identity?

Solution.

No. It holds exactly when $A \subseteq B$, not for all sets. For $A = \{1\}$, $B = \emptyset$, the left side is $\emptyset$ while the right side is $\{1\}$.

HARD WORKED EXAMPLE - Find exact validity condition

Problem. For which A, B does $A - B = A$ hold?

Solution.

Exactly when $A \cap B = \emptyset$. If disjoint, B removes nothing from A . Conversely, if $A - B = A$, no A -element belongs to B , so $A \cap B = \emptyset$.

17. Counterexamples to universal claims

DEFINITION - Counterexamples to universal claims

A counterexample is one legal assignment of sets or elements that makes a universal statement false.

Choose values from a region where the proposed sides differ. Then compute both sides explicitly and display the separating element.

Minimal counterexamples are usually clearest: U with one or two elements often suffices.

EXAM NOTE - Do not merely say “false”

State U and all sets, calculate LHS and RHS, then point out the mismatch.

MEDIUM WORKED EXAMPLE - Disprove a false disjointness claim

Problem. Disprove: $A \cap B = \emptyset$ implies $A = \emptyset$ or $B = \emptyset$.

Solution.

Take $U = \{1, 2\}$, $A = \{1\}$, $B = \{2\}$. Then $A \cap B = \emptyset$, but both A and B are non-empty.

HARD WORKED EXAMPLE - Target a failed symmetric-difference law

Problem. Disprove $(A+B) \cup C = (A \cup C) + (B \cup C)$ for all A, B, C .

Solution.

Let $U = \{c\}$, $A = B = \emptyset$, $C = \{c\}$. $LHS = (\emptyset + \emptyset) \cup C = \{c\}$. $RHS = C + C = \emptyset$. Hence the universal claim is false.

18. Proof or counterexample workflow

DEFINITION - Proof or counterexample workflow

First investigate the claim; then choose the correct final method.

1. Parse quantifiers and operators. 2. Compare regions or membership conditions. 3. If equivalent, prove by arbitrary-element logic. 4. If not, build a counterexample.

Never force an algebraic proof after discovering a region where the two sides differ.

DEFINITION - Decision rule

A universal equality needs a proof if true and one counterexample if false.

MEDIUM WORKED EXAMPLE - Decide a claim

Problem. Is $X - (Y \cup Z) = (X - Y) \cap (X - Z)$ universally true?

Solution.

Yes. Region testing gives the same X-only-with-respect-to-Y-and-Z region, and the iff proof follows De Morgan as shown earlier.

HARD WORKED EXAMPLE - Diagnose a false claim

Problem. Is $X - (Y - Z) = (X - Y) - Z$ universally true?

Solution.

No. Let $U = \{a\}$, $X = Z = \{a\}$, $Y = \emptyset$. $LHS = X - \emptyset = \{a\}$; $RHS = (X - \emptyset) - Z = \emptyset$. The element a separates the sides.

19. Power-set containment and element type

DEFINITION - Power-set containment and element type

If $X \in P(A)$, then X is a set and $X \subseteq A$.

Power-set proofs operate one level above ordinary membership. The arbitrary “element” of $P(A)$ is itself a subset.

$$X \in P(A \cup B) \text{ iff } X \subseteq A \cup B$$

COMMON TRAP - Type warning

Do not replace $X \in P(A)$ with $X \in A$. The first says X is a subset of A ; the second says X is an element of A .

MEDIUM WORKED EXAMPLE - Verify a containment

Problem. Let $A = \{1\}$, $B = \{2\}$. List $P(A) \cup P(B)$ and show it lies in $P(A \cup B)$.

Solution.

$P(A) \cup P(B) = \{\emptyset, \{1\}, \{2\}\}$. Each is a subset of $\{1, 2\}$, so each belongs to $P(A \cup B)$.

HARD WORKED EXAMPLE - Show containment is proper

Problem. With $A = \{1\}$, $B = \{2\}$, prove $P(A) \cup P(B) \subset P(A \cup B)$.

Solution.

Containment holds generally. But $\{1, 2\} \in P(A \cup B)$ and $\{1, 2\} \notin P(A) \cup P(B)$. Thus the containment is strict.

20. Power-set intersection identity

DEFINITION - Power-set intersection identity

$$P(A) \cap P(B) = P(A \cap B).$$

A set X lies in both power sets exactly when every element of X belongs to both A and B .

This is an equality, unlike the union situation.

EXAM NOTE - Work one level higher

The proof variable X is a set, not an ordinary point x .

MEDIUM WORKED EXAMPLE - Concrete verification

Problem. Let $A = \{1, 2\}$, $B = \{2, 3\}$. Compute both sides.

Solution.

$P(A) \cap P(B) = \{\emptyset, \{2\}\}$. Since $A \cap B = \{2\}$, $P(A \cap B) = \{\emptyset, \{2\}\}$.

HARD WORKED EXAMPLE - Formal proof

Problem. Prove $P(A) \cap P(B) = P(A \cap B)$.

Solution.

$X \in P(A) \cap P(B)$ iff $X \in P(A)$ and $X \in P(B)$ iff $X \subseteq A$ and $X \subseteq B$ iff $X \subseteq A \cap B$ iff $X \in P(A \cap B)$.

21. The inclusion-exclusion principle

DEFINITION - The inclusion-exclusion principle

For finite sets A and B , $|A \cup B| = |A| + |B| - |A \cap B|$.

Adding $|A|$ and $|B|$ counts each common element twice. Subtracting the intersection once leaves every union element counted exactly once.

The formula is a counting theorem, not an unexplained arithmetic trick.

NOTE - Why subtraction appears

The overlap is included once in $|A|$ and once in $|B|$, so one duplicate copy must be removed.

MEDIUM WORKED EXAMPLE - Apply the theorem

Problem. If $|A|=24$, $|B|=19$ and $|A \cap B|=7$, find $|A \cup B|$.

Solution.

$$|A \cup B| = 24 + 19 - 7 = 36.$$

HARD WORKED EXAMPLE - Recover an unknown intersection

Problem. If $|A \cup B|=70$, $|A|=46$ and $|B|=39$, find $|A \cap B|$.

Solution.

$$70 = 46 + 39 - |A \cap B|, \text{ so } |A \cap B| = 85 - 70 = 15.$$

22. The sum rule for disjoint sets

DEFINITION - The sum rule for disjoint sets

If finite A and B are disjoint, then $|A \cup B| = |A| + |B|$.

Disjointness means $A \cap B = \emptyset$, so $|A \cap B| = 0$ in inclusion-exclusion.

The sum rule is therefore a special case, not a separate unrelated fact.

COMMON TRAP - Check disjointness first

Direct addition is invalid when an overlap may exist.

MEDIUM WORKED EXAMPLE - Use the sum rule

Problem. A has 13 elements, B has 21, and $A \cap B = \emptyset$. Find $|A \cup B|$.

Solution.

$$|A \cup B| = 13 + 21 = 34.$$

HARD WORKED EXAMPLE - Partition argument

Problem. Suppose U is partitioned into pairwise disjoint A, B, C with sizes 17, 23, 11. Find $|U|$ and justify.

Solution.

Every U -element lies in exactly one part, so no overlap is counted twice. Repeated sum rule gives $|U| = 17 + 23 + 11 = 51$.

23. Two-set survey problems

DEFINITION - Two-set survey problems

Translate “A or B” as $A \cup B$, “both” as $A \cap B$, “only A” as $A - B$, and “neither” as $(A \cup B)'$.

Fill the overlap first, then subtract it from each total. Finally compare the union with $|U|$.

only A = $|A| - |A \cap B|$; neither = $|U| - |A \cup B|$

EXAM NOTE - Words are operators

“Or” is inclusive unless the question explicitly says “but not both”.

MEDIUM WORKED EXAMPLE - Group of learners

Problem. Among 50 learners, 25 play game M, 30 play sport B and 10 do both. Find the union and neither.

Solution.

$|M \cup B| = 25 + 30 - 10 = 45$. Therefore neither = $50 - 45 = 5$.

HARD WORKED EXAMPLE - Reconstruct every region

Problem. Of 1000 students, 700 take M, 400 take C and 800 take M or C. Find both, only M, only C and neither.

Solution.

Both = $700 + 400 - 800 = 300$. Only M = 400; only C = 100; neither = $1000 - 800 = 200$. The four regions sum to 1000.

24. Three-set inclusion-exclusion

DEFINITION - Three-set inclusion-exclusion

For finite A, B, C , add singles, subtract pairwise intersections, then add the triple intersection.

The triple region is first counted three times, then removed three times by pairwise subtraction, leaving zero; adding it once restores the correct count.

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

COMMON TRAP - Pair totals include the centre

Convert each pairwise count to “pair only” by subtracting the triple intersection.

MEDIUM WORKED EXAMPLE - Calculate a union

Problem. Given sizes 30, 25, 20; pair intersections 8, 6, 5; triple 2. Find the union.

Solution.

$$30 + 25 + 20 - 8 - 6 - 5 + 2 = 58.$$

HARD WORKED EXAMPLE - Find exactly one

Problem. Using the same data, find how many belong to exactly one set.

Solution.

$$\text{Only } A = 30 - (8 - 2) - (6 - 2) - 2 = 18. \text{ Only } B = 25 - 6 - 3 - 2 = 14. \text{ Only } C = 20 - 4 - 3 - 2 = 11. \text{ Exactly one} = 43.$$

25. Solving an unknown central intersection

DEFINITION - Solving an unknown central intersection

When the triple overlap is unknown, call it x and express every surrounding region in terms of x .

Use either the three-set formula or the fact that all elementary regions plus the outside region equal $|U|$.

After solving, verify every derived region is a non-negative integer.

EXAM NOTE - Algebra check

An impossible negative region signals inconsistent data or a mistranslated “only”.

MEDIUM WORKED EXAMPLE - Machine operators

Problem. Exactly 45 workers can operate at least one of crane C, backhoe B, bulldozer D. Given $|C|=22$, $|B|=26$, pair intersections 8,4,6 and triple 1, find $|D|$.

Solution.

$45 = 22 + 26 + |D| - 8 - 4 - 6 + 1 = 31 + |D|$, so $|D| = 14$.

HARD WORKED EXAMPLE - Software techniques

Problem. All 22 engineers use at least one of F,U,E. Given $|F|=17$, $|U|=9$, $|E|=9$; pair intersections 5,4,7. Find the triple intersection and U-only.

Solution.

Let triple = x . Then $22 = 17 + 9 + 9 - 5 - 4 - 7 + x = 19 + x$, so $x = 3$. U-only = $9 - (5 - 3) - (7 - 3) - 3 = 0$.

26. Proofs involving specific sets

DEFINITION - Proofs involving specific sets

For concrete set-builder sets, translate membership into algebraic conditions and prove those conditions equivalent.

Variable names inside set-builder notation are placeholders. The proof may use one fresh arbitrary variable x throughout.

Factoring, sign analysis and domain restrictions must all be reversible in an iff proof.

EXAM NOTE - Domain changes answers

A real inequality may define an interval; the same inequality over integers defines only the integers inside that interval.

MEDIUM WORKED EXAMPLE - Quadratic interval proof

Problem. Prove $\{x \in \mathbb{R} \mid x^2 + 6x + 5 < 0\} = \{x \in \mathbb{R} \mid -5 < x < -1\}$.

Solution.

Factor $(x+5)(x+1) < 0$. The product is negative exactly when the factors have opposite signs, which for ordered roots $-5 < -1$ occurs exactly for $-5 < x < -1$.

HARD WORKED EXAMPLE - Integer admission proof

Problem. Prove $\{x \in \mathbb{Z} \mid x^2 - 5x + 4 < 0\} = \{x \in \mathbb{Z}^+ \mid x \text{ is a prime factor of } 6\}$.

Solution.

$(x-1)(x-4) < 0$ iff $1 < x < 4$. With $x \in \mathbb{Z}$, this gives $\{2, 3\}$. The positive prime factors of 6 are exactly 2 and 3, so both descriptions define the same set.

27. Implications, transitivity and existence claims

DEFINITION - Implications, transitivity and existence claims

A statement with “if” needs a forward proof; a statement with “there exists” needs a witness; a universal existence claim must work for every allowed starting set.

Subset is transitive: $X \subseteq Y$ and $Y \subseteq Z$ imply $X \subseteq Z$. Union and intersection are monotone: $V \subseteq W$ preserves containment after union or intersection with Z .

Existence claims often fail at edge cases such as $X \neq \emptyset$ or $X \neq U$.

EXAM NOTE - Match proof to quantifier

For “for every X there exists Y ”, Y may depend on X , but it must exist for every X .

MEDIUM WORKED EXAMPLE - Prove monotonicity

Problem. If $V \subseteq W$, prove $\forall n Z \subseteq W \cap Z$ and $\forall u Z \subseteq W \cup Z$.

Solution.

If $x \in V \cap Z$, then $x \in V \subseteq W$ and $x \in Z$, hence $x \in W \cap Z$. If $x \in V \cup Z$, then $x \in V$ or $x \in Z$; in the first case $x \in W$, so in either case $x \in W \cup Z$.

HARD WORKED EXAMPLE - Analyse two existence claims

Problem. Is it true that for every $X \subseteq U$ there exists $Y \subseteq U$ with (i) $X \cup Y = \emptyset$ and (ii) $X \cap Y = U$?

Solution.

(i) False: if $X = \{a\}$, every union contains a . It works only for $X = \emptyset$ with $Y = \emptyset$. (ii) False unless $X = U$: an intersection is a subset of X , so it cannot equal U when $X \subset U$. If $X = U$, choose $Y = U$.

Master proof-method selector

Choose the proof method from the logical form of the claim, not from which technique feels familiar.

CLAIM FORM	BEST METHOD	WHAT MUST APPEAR
$S=T$ for all sets	Iff chain or mutual containment	Translate arbitrary membership both ways.
$S \subseteq T$ for all sets	One-direction element proof	Let $x \in S$ and derive $x \in T$.
Universal claim appears false	Counterexample	Give legal sets, compute both sides, show mismatch.
Specific finite sets equal	Roster comparison or both containments	List all distinct elements or prove membership equivalence.
Specific set-builder sets equal	Algebraic iff chain	Translate admission rules and keep every step reversible.
Finite cardinality with overlap	Inclusion-exclusion	Correct repeated counting of intersections.
Existence claim	Construct a witness	Give an allowed object and verify every required property.

Equality-proof template

Let x be arbitrary. $x \in \text{LHS}$ iff ...definitions and logic... iff $x \in \text{RHS}$. Therefore $\text{LHS}=\text{RHS}$. QED

Counterexample template

Let $U=\dots$, $A=\dots$, $B=\dots$. Then $\text{LHS}=\dots$ while $\text{RHS}=\dots$. Since witness w lies in one side but not the other, $\text{LHS} \neq \text{RHS}$.

EXAM NOTE - No skipped bridge

Whenever a proof changes a set expression into words such as “and”, “or” or “not”, name the definition or logical equivalence being used.

Exam traps and mark-saving checks

COMMON TRAP - Diagram is not the final proof

Use matching shading as evidence, then write the membership proof when asked to prove.

COMMON TRAP - Only one direction

Showing $LHS \subseteq RHS$ does not prove $LHS = RHS$. You still need $RHS \subseteq LHS$.

COMMON TRAP - Broken iff chain

Every iff step must reverse. If you only know implication, use $\Rightarrow$ and do not claim equivalence.

COMMON TRAP - Weak counterexample

Random sets that happen to satisfy a false universal claim prove nothing. Target a region where the proposed expressions differ.

COMMON TRAP - Power-set type error

$X \in P(A)$ means $X \subseteq A$. The variable X represents a set.

COMMON TRAP - Pair intersections

In a three-set problem, $|A \cap B|$ normally includes the triple intersection. Subtract the centre to get A-and-B only.

COMMON TRAP - Missing universe

Complements and outside regions require U . State it.

Final checklist

- Did I identify whether the claim is universal, existential or about specific sets?
- For equality, did I prove both directions?
- Is every iff step genuinely reversible?
- If false, did I calculate both sides of one valid counterexample?
- In a power-set proof, did I keep the variable at the correct set level?
- In counting, do all elementary regions add to $|U|$ and remain non-negative?

Final integrated challenge

This combines diagram reasoning, identity proof, counterexample design, power sets and three-set counting.

DEFINITION - Given

Let $A, B, C \subseteq U$ be finite. For the counting part, $|U|=90$, $|A|=44$, $|B|=38$, $|C|=35$, $|A \cap B|=16$, $|A \cap C|=14$, $|B \cap C|=13$ and $|A \cap B \cap C|=6$.

Questions

- 1. Prove $A - (B \cup C) = (A - B) \cap (A - C)$.
- 2. Decide whether $A - (B - C) = (A - B) - C$ is an identity; prove or disprove.
- 3. Prove $P(A \cap B) = P(A) \cap P(B)$.
- 4. Find $|A \cup B \cup C|$, the number in exactly A, and the number outside all three.
- 5. Explain why matching Venn shading alone is not the strongest final proof.

Fully worked solution

1. For arbitrary x : $x \in A - (B \cup C)$ iff $x \in A$ and $x \notin B \cup C$ iff $x \in A$, $x \notin B$ and $x \notin C$ iff $x \in A - B$ and $x \in A - C$ iff $x \in (A - B) \cap (A - C)$.
2. It is false. Let $U = \{a\}$, $A = C = \{a\}$, $B = \emptyset$. Then $A - (B - C) = \{a\}$, while $(A - B) - C = \emptyset$. The witness a separates the sides.
3. For arbitrary set X : $X \in P(A \cap B)$ iff $X \subseteq A \cap B$ iff $X \subseteq A$ and $X \subseteq B$ iff $X \in P(A)$ and $X \in P(B)$ iff $X \in P(A) \cap P(B)$.
4. $|A \cup B \cup C| = 44 + 38 + 35 - 16 - 14 - 13 + 6 = 80$. Exactly $A = 44 - (16 - 6) - (14 - 6) - 6 = 20$. Outside all three $= 90 - 80 = 10$.
5. A generic Venn diagram assumes a region layout and becomes difficult beyond three sets. Element proof uses definitions and covers empty, coincident and other edge cases without relying on appearance.

EXAM NOTE - Integrity check

Pair-only values are 10, 8 and 7; the centre is 6; all regions are non-negative and the union plus outside equals 90.

Closed-book exit test

Attempt every item without notes, then mark it from the memorandum.

- 1. Define set identity and counterexample.
- 2. State the mutual-containment test for equality.
- 3. Prove $(A \cup B)' = A' \cap B'$ using iff.
- 4. Prove $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.
- 5. Disprove: $A \cap B = \emptyset$ implies $A = \emptyset$ or $B = \emptyset$.
- 6. Explain the type of the variable X in $X \in P(A)$.
- 7. If $|A|=31$, $|B|=27$ and $|A \cup B|=49$, find $|A \cap B|$.
- 8. Three sets have sizes 20,18,16; pair intersections 7,6,5; triple 2. Find the union.
- 9. Prove $\{x \in \mathbb{R} \mid x^2 + x - 2 > 0\} = \{x \in \mathbb{R} \mid x < -2 \text{ or } x > 1\}$.
- 10. State one limitation of Venn-diagram proofs.

Concise memorandum

1. Identity: true for every allowed value of the variables. Counterexample: one allowed choice making a universal claim false.
2. $A=B$ iff $A \subseteq B$ and $B \subseteq A$.
3. $x \in (A \cup B)'$ iff $x \notin A \cup B$ iff $x \notin A$ and $x \notin B$ iff $x \in A' \cap B'$.
4. Translate to $x \in A$ and $(x \in B \text{ or } x \in C)$, then distribute “and”.
5. $U=\{1,2\}$, $A=\{1\}$, $B=\{2\}$; the intersection is empty but both sets are non-empty.
6. X is itself a set, and $X \in P(A)$ iff $X \subseteq A$.
7. $31+27-49=9$.
8. $20+18+16-7-6-5+2=38$.
9. Factor $(x+2)(x-1)>0$; the factors have the same sign outside the roots.
10. It becomes unwieldy beyond three sets and visual regions may wrongly appear non-empty.